

MATH 202 - DIFFERENTIAL EQUATIONS SYLLABUS – SPRING 2026

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Office Hours: Monday 9:40-10:30, or by appointment.

Lectures. Monday 8.40-9.30, Wednesday 14.40-16.30. FENS G077

Recitations. Sec A Wed 16.40-17.30 Sec B Wed 17.40-18.30 Sec C Thu 17.40-18.30 Sec D M 9.40-10.30.

Course Description. Differential equations are used to model the behavior of systems in the natural world, such as population growth and decay, spring-mass systems or electrical circuits, and predict how these systems will behave in the future. This course will present the basics of solving differential equations.

We will cover the topics First-order ordinary differential equations and solution methods. Direction fields, qualitative methods. First order, second-order linear and higher order ordinary differential equations. Mathematical models. Laplace transform. Linear systems.

Goals. The purpose of the course is to give knowledge of how to solve ordinary differential equations. Upon completing this course, students should understand the general theory of ordinary differential equations and the basic techniques for solving differential equations involving one unknown function and one independent variable.

Text Book. (look at SuCourse for more info on textbook!) Fundamentals of Differential Equations and Boundary Value Problems 6th edition, R. K. Nagle, E. B. Saff, A. D. Snider, Pearson Education International, 2012

Recitations. The lectures in the recitation classes will be in discussion format. In the recitations only some selected problems will be discussed. There will **not** be a review of the course material. Make sure to study the lecture notes before coming to the recitations. You must try the problems at the end of each section in the book.

Exams. See “Grading” for details. **It is your responsibility to be present in the exams.** The tentative date of the midterm is below.

Makeup Policy. If you miss an exam and wish to make it up, you must contact the instructor N. G. Göğüş by email, and explain your excuse as soon as possible. You cannot take makeup exam for both midterm and the final exam. Only students that have contacted the instructor with a valid excuse will be contacted to arrange the terms of the exam. If it is a health problem, you must bring a medical report, which must be given or checked by the SU Health Center within 3 days of the end date of the report.

Academic Honesty. The strength of the university depends on academic and personal integrity. In this course, you must be honest and truthful. Cheating hurts our community by undermining academic integrity, creating mistrust, and fostering unfair competition. The university will punish cheaters with failure on an assignment, failure in a course, permanent transcript notation, suspension, and/or expulsion. Violations can include cheating on exams, plagiarism, reuse of assignments without permission, improper use of the Internet and electronic devices unauthorized collaboration, alteration of graded assignments, forgery and falsification, lying, facilitating academic dishonesty, and unfair competition. Ignorance of these rules is not an excuse.

In this course, as in many math courses, working in groups to study particular problems and discuss theory is strongly encouraged. Your ability to talk mathematics is of particular importance to your

general understanding of mathematics. You should collaborate with other students in this course on the general construction of exercise problems. However, you must write up the solutions to these problems individually and separately. If there is any question as to what this statement means, please see the professor.

AI tools and other open sources

Please note that improper use of GenAI tools is a serious breach of academic integrity. Violations of academic integrity result in disciplinary action. For more information, refer to the [Sabancı University Academic Integrity Handbook for Students](#) on the CIAD webpage.

Attendance. Attendance at lectures and recitations is encouraged, however, it will not be followed. If you miss a class, it is your responsibility to make up the material.

Grading: There will be two midterm exams (30 points each) and one final exam (40 points).

The dates of the midterms may change!

Mid 1	25 March 2026 Wednesday	30%
Mid 2	29 April 2026 Wednesday	30%
Final	To be determined by SR	40%

The grading scale is **89-100 A, 84-88 A-, 79-83 B+, 74-78 B, 69-73 B-, 64-68 C+, 57-63 C, 53-56 C-, 49-52 D+, 41-48 D, and below 41 F. Plus and minus grades will be given at my discretion.**

Sec	Name of TA	Lecture Time	Office hrs
A1		Wed 16.40-17.30	
A2		Wed 16.40-17.30	
A3		Wed 16.40-17.30	
B1		Wed 17.40-18.30	
B2		Wed 17.40-18.30	
C1		Thu 17.40-18.30	
C2		Thu 17.40-18.30	
D1		Monday 9.40-10.30	
D2		Monday 9.40-10.30	

Tentative Course Schedule:

Weeks	Sections	
1	1.1, 1.2, 2.1	
2	2.2, 2.3, 2.4	
3	2.5, 2.6 some parts omitted	
4	3.1, 3.2, 3.3, 3.4	
5	4.1, 4.2, 4.3	
6	4.4, 4.5, 4.6	
7	4.7, 6.1, 6.2, 6.3	
8	7.1, 7.2, 7.3, 7.4, 7.5	
9	7.6, 7.7, 7.9	
10	9.1, 9.2	
11	9.3, 9.4	
12	9.5	
13	9.6, 9.8	
14	Review	

Holidays

19-22 March Thur-Sun Ramazan Bayramı

23 April Thur

1 May Fri

19 May Tue

25-30 May Mon-Sat Kurban Bayramı

Last day of classes: 22 May Fri

Final Exams: 1-10 June

Makeup Exams: 11-13 June

Add-drop: 26-27 Feb

Withdrawal: 30 Mar-24 Apr

Content of the exams. Roughly the following topics will be the content of the exams.

Midterm 1 25 March 2026 Wednesday 14.40-16.30 lecture time.

1.1, 1.2 Solutions of initial value problems.

2.1, 2.2, 2.3 Separable equations, linear first order equations.

2.4, 2.5, 2.6 Exact equations, integrating factors, substitutions and transformations.

3.1, 3.2, 3.3, 3.4 Modeling problems involving first order differential equations, radioactive decay, population models and Newtonian mechanics.

4.1, 4.2, 4.3 Mass-spring example, general solution of a homogeneous second order linear equation, auxiliary equations with complex roots.

Midterm 2 29 April 2026 Wednesday 14.40-16.30 lecture time.

4.4, 4.5, 4.6, 4.7 Undetermined coefficients, variation of parameters, variable-coefficient equations.

6.1, 6.2, 6.3 Higher order differential equations

7.1, 7.2, 7.3 Laplace transform: Definition and properties.

7.4, 7.5, 7.6 Inverse Laplace transform, solving IVPs.

Final All topics above plus the following

7.7 Convolution.

9.1, 9.2, 9.3 Review of linear algebra.

9.4, 9.5, 9.6, 9.8 Linear systems in normal form, homogeneous linear systems with constant coefficients, complex eigenvalues, matrix exponential function.

Important!!

Some information in the syllabus may change within the first weeks of the semester.

If you need extra help:

Always keep in mind that it is very natural and all right if you do not understand a concept or something in the lecture. I and teaching assistants are available to help. Ask questions. Benefit office hours.

- **Regular study habits are sufficient to get a decent grade.**
- **Attend the classes and recitation hours regularly.** Make sure you attend your own (registered) recitation section.
- Studying out of class for this course should become a routine. Key to success in mathematics is practice. **Solve many problems** related to each concept.
- **You are responsible from every announcement made in class or in SUCourse.** Not attending the class or not following SUCourse regularly is not an excuse, in case you miss something.
- **I am available to help.** Feel free to ask me and your TA a question in and out of class. If you cannot make it to our office hours, you should e-mail me (or your TA) ahead of time and make an appointment.